VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	22BAI1266

Continuous Assessment Test (CAT)–2 MARCH 2025

Programme	:	B.Tech. (Artificial Intelligence and Machine Learning)	Semester	:	Winter 2024-25
Course Code & Course Title	:	BCSE417L-Machine Vision	Class Number	:	CH2024250502402 CH2024250502405
Faculty	:	Dr Bharadwaja Kumar Dr S.Rajarajeswari	Slot	:	A1+TA1
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.

Answer all questions

Q. No	Sub Sec.	Description	Marks
1		Suppose that you have applied Laplacian kernel for edge detection and found that there are many false positive edges caused by noise.	
	a)	Propose an optimized multi-step approach to mitigate noise artifacts in edge detection while ensuring accurate edge preservation. Justify your methodology with a mathematical basis for noise reduction and edge enhancement. **(4 Marks)**	
	b)	Develop any discrete kernel with $\sigma = 1.0$ using the proposed methodology and apply it on the marked pixel in the image shown below **(6 Marks)**	10

170	238	85	255	221	0
68	136	17	170	119	68
221	0	238	136	0	255
119	255	85	170	136	238
238	17	221	68	119	255
85	170	119	221	17	136

2		A brain MRI scan applies region-growing segmentation to detect tumors. The algorithm starts at a seed pixel with intensity 150 and expands in 8 directions based on intensity similarity and find the region Initially, a fixed threshold of ±15 of the seed is used for region segmentation. Due to intensity variations, adaptive thresholding is introduced using the formula: $T = I_{region} - 0.1 \times I_{region}$	10

I_{region} is the mean intensity of the neighbours(using 8 connectivity) of the current pixel

Demonstrate the step wise process in both fixed and adaptive thresholding-based region segmentation and compare them.
Given the grayscale intensity matrix:

140	155	160	145
148	150	152	165
135	158	162	170
130	145	155	175

3 A satellite imaging company is analyzing texture patterns in a grayscale aerial image of a city to classify different land types such as urban areas, water bodies, and vegetation. The image is represented as an **8×8** grayscale matrix with pixel intensity values ranging from **0 to 5** (normalized gray levels).
To enhance classification accuracy, the company decides to analyze how pixel intensities co-occur at two different spatial relationships in the image:

0	1	2	3	2	1	4
1	2	3	4	0	1	1
2	3	4	0	1	2	2
3	4	0	1	2	3	3
4	3	1	2	3	4	4
1	2	2	3	4	0	3

Analyze the occurrence of pixel pairs positioned right with a spatial distance of 1 and pixel pairs positioned along the left-up diagonal with a spatial separation of 2. Compare both the representation. **10**

4 Determine the one-pixel-wide skeleton of the below given image matrix using the given structuring element(by rotating 180°) , and then compute the second skeleton in the same way of the given image matrix in the same way as above with the "don't care" values as 1. Compare both skeletons based on the number of pixels removed from the original image.

Structuring element is **10**

X	1	X
0	1	0
X	1	X

Image Matrix is

0	0	0	1	1	1	0	0
0	1	1	1	1	1	1	0
1	1	0	1	1	1	0	1
1	1	1	0	1	0	1	1
1	1	1	0	1	0	1	1
0	1	1	1	1	1	0	1
0	0	1	1	0	0	1	1
1	1	1	0	0	0	0	1

5 To analyze road signs, lane markings, and obstacles clearly on a self-driving car , the engineers must determine the best processing approach among Spatial domain processing and Frequency domain processing .

a) Examine the distinctions and similarities between image processing methods that manipulate pixel intensity values directly and those that modify an image by analyzing its frequency components **(4 Marks)**

b) Given a 4×4 image matrix, apply a mathematical transformation to extract its frequency components and then reconstruct the original image. Demonstrate how this process enhances critical details while maintaining computational efficiency. **(6 Marks)** 10

80	0	70	0
10	20	30	40
50	12	24	36
53	55	21	33

**********All the best **********